

6 Digit Nixie Clock "64Bit" - Driver Board

64Bit comes from the fact that this clock is driven with shift registers. Between the 6 nixie digits, and indicators, 64 bits are sent from the microcontroller to the shift registers to create the display output

This is an STM32F446 based project.

The clock time keeping is handled via a Real Time Clock (RTC) communicated with via I2C. The STM32 also has an on-board RTC that can be utilized. Both RTCs have a battery backup via a replaceable 2032 coin cell.

The interface with the clock is via a rotary encoder with a push button. Interfacing with remote capacitive buttons is possible via connectors on the bottom of the Driver Board.

There is an ESP32-C6 module driving a full-color LCD screen. The ESP32 features wifi and bluetooth allowing for automatic time setting and smart home integration. Communication with the STM32 is via UART.

This clock was designed with safety in mind. There is a dedicated safety circuit that cuts the input and output connections of the high voltage power supply via relays. There are connectors for interfacing external spring pin connections to the cover of the clock, ensuring the high voltage power supply is not on while the lid is off. The safety circuit features dual channels and feedback to the microcontroller. With the spring pin feature, it will provide isolation from the high voltage, even in an event where the microcontroller experiences an error.

The power design of the clock relays on simple components. There are two inputs available to power the clock. A USB-C port and a 12V barrel jack. A boost converter utilizing XL6009 converts (boost) the incoming voltage to 12VDC. Then an LDO creates 5V then another LDO, in series, creates 3.3V. The high voltage power supply typically is set to output 170VDC to 180VDC. It is supplied by a power supply module that sources 12V power.

This driver board interfaces with the display board PCB via right angle board-to-board connectors. Mounted to the back of the display board is an LED board to provide RGB backlighting to each IN-12 Nixie Tubes.

The circuit boards will be mounted to a 3D printed base via brass standoffs. Overtop of the PCBs will be a glass top. The glass top will have sections where copper tape is applied. Spring pins contact the copper tape to pass connections for the safety circuit.

STM32 MCU

I2C

Interface

Safety Circuit

Power

Connectors

Connectors 2

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Sheet: /

File: 64 Bit - Main Board.kicad_sch

Title: Project Overview

Size: A4

Date: 2025-12-01

Rev: 1

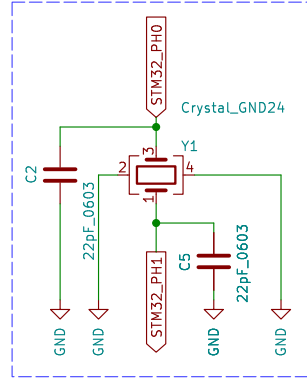
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Id: 1/9

STM32F446 Processor ARM M4
168MHz

Level Shifting

main crystal



ambient light sensing

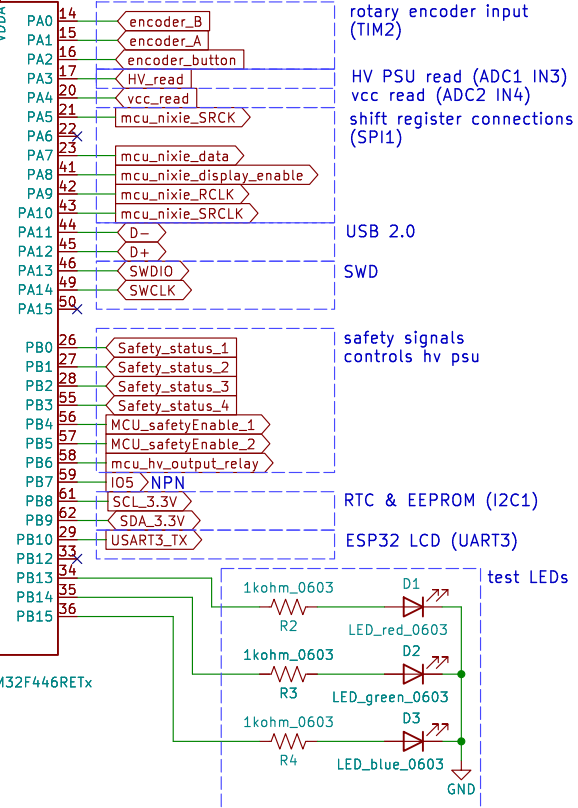
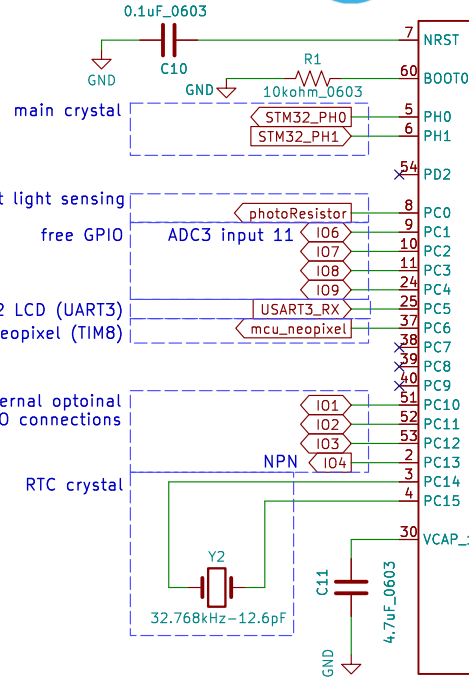
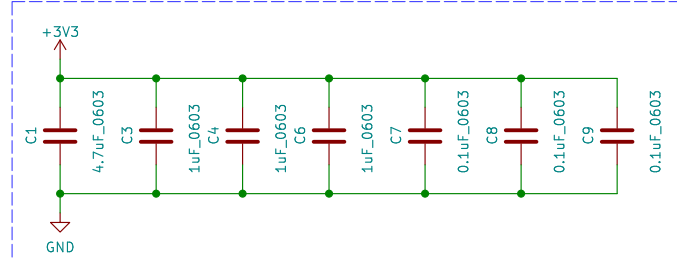
free GPIO

ESP32 LCD (UART3)
neopixel (TIM8)

external optional
GPIO connections

RTC crystal

STM32 bypass caps



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Sheet: /STM32 MCU/
File: STM32 MCU.kicad_sch

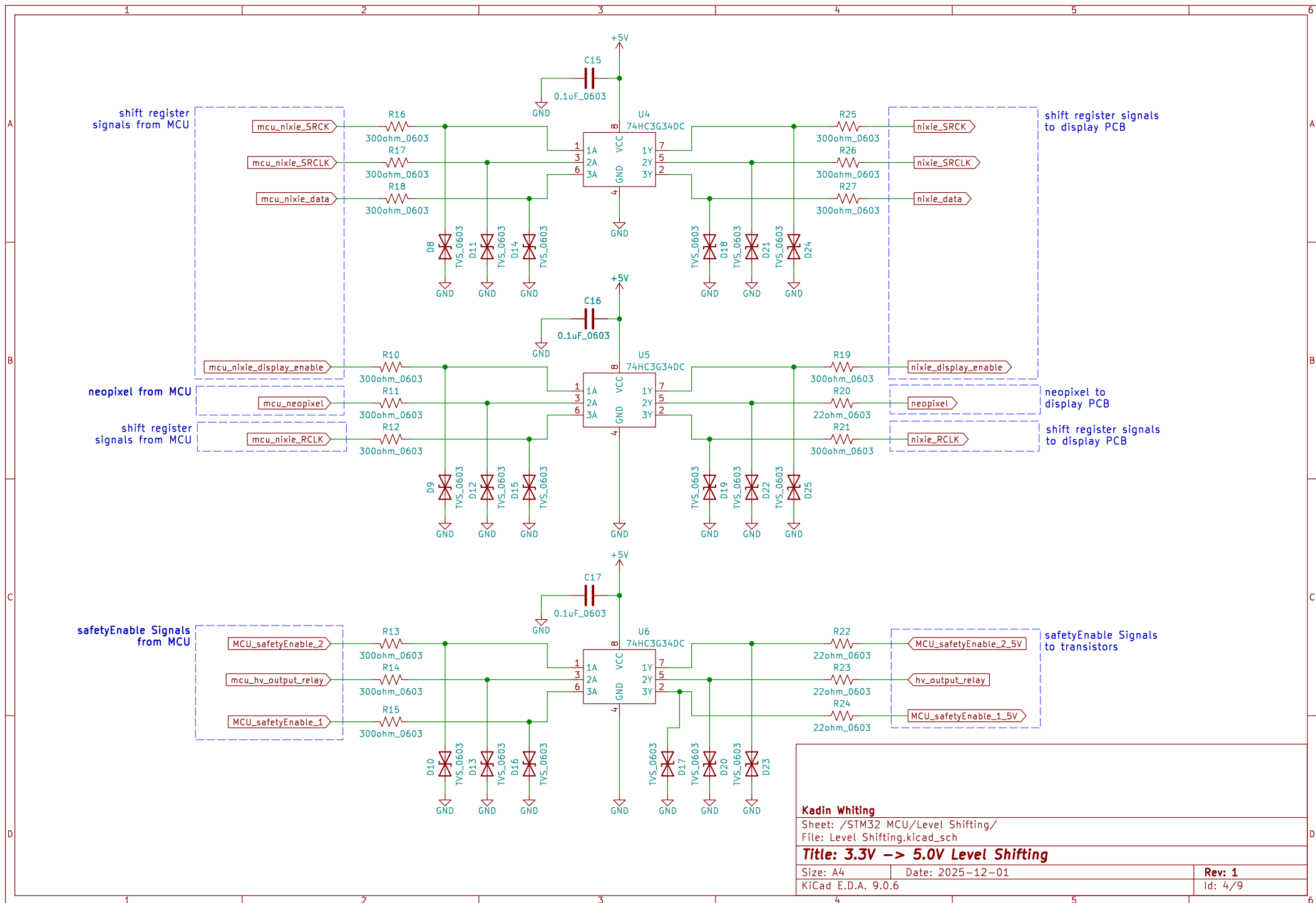
Title: Microcontroller

Size: A4 Date: 2025-12-01

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Rev: 1

Id: 2/9



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Sheet: /STM32 MCU/Level Shifting/
File: Level Shifting.kicad_sch

Title: 3.3V -> 5.0V Level Shifting

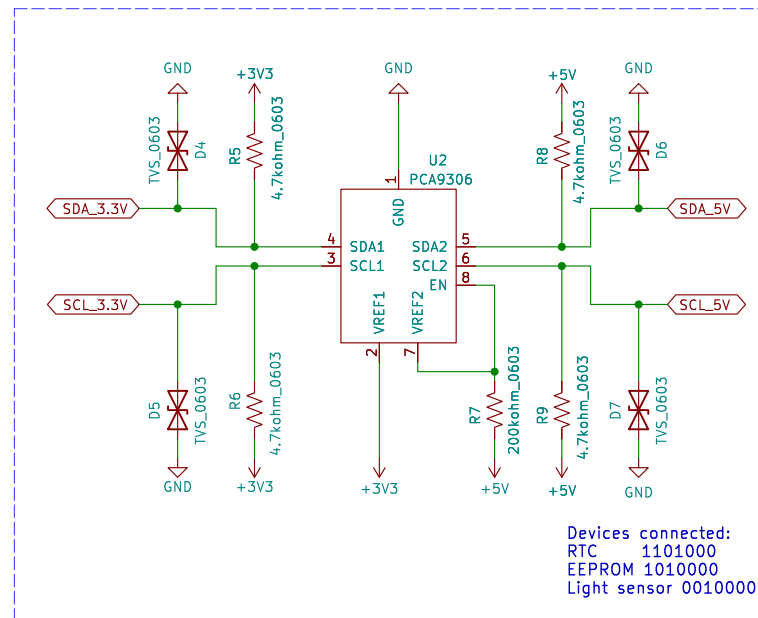
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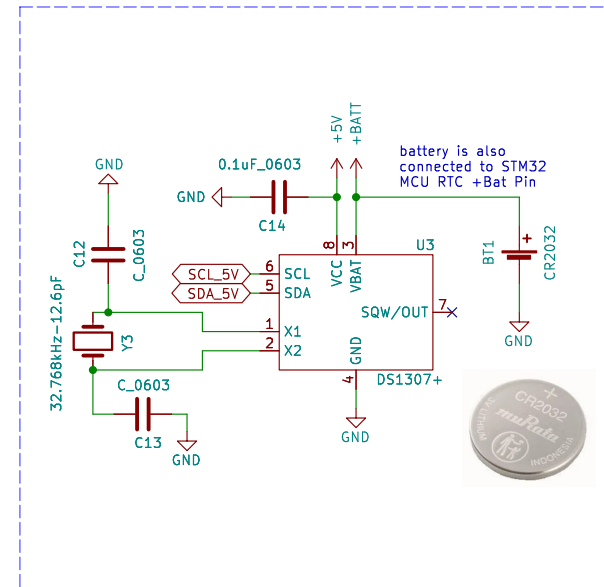
Rev: 1

Id: 4/9

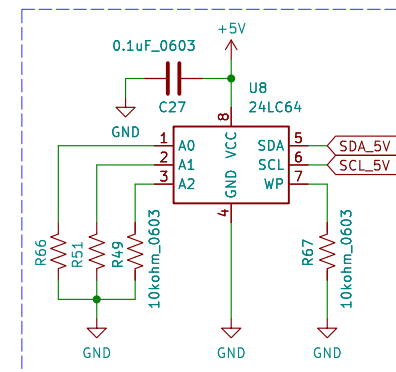
3.3V to 5V I2C level shifter



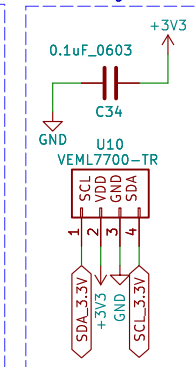
Real Time Clock



EEPROM



ambient light sensor



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Sheet: /I2C/
 File: I2C.kicad_sch

Title: RTC & RTC Level Shifting

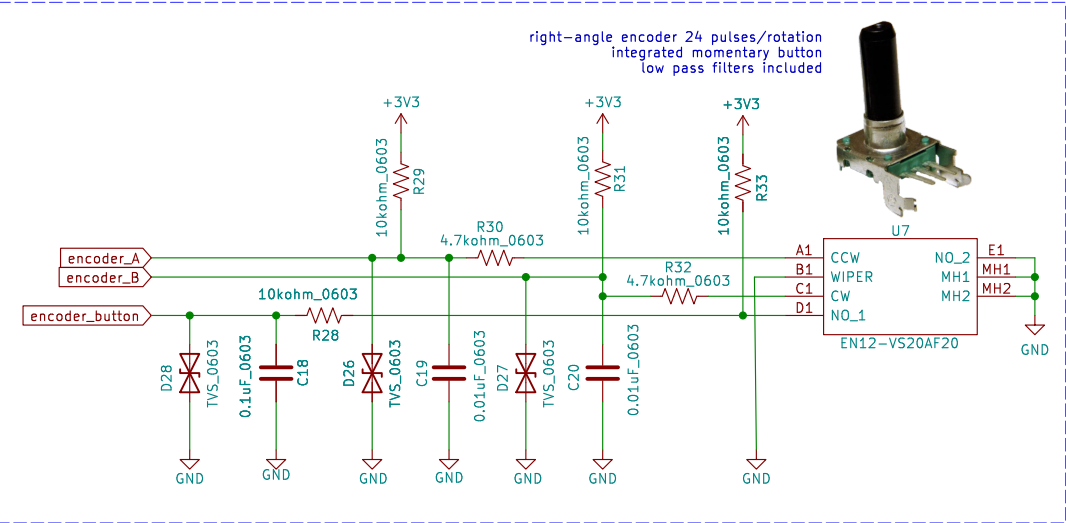
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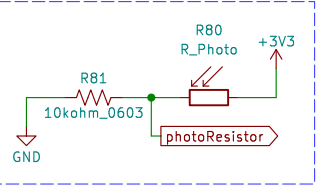
Rev: 1

Id: 3/9

rotary encoder with push button



photoresistor
ambient light sensing



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Sheet: /Interface/
File: Interface.kicad_sch

Title: Interface – Rotary Encoder

Size: A4

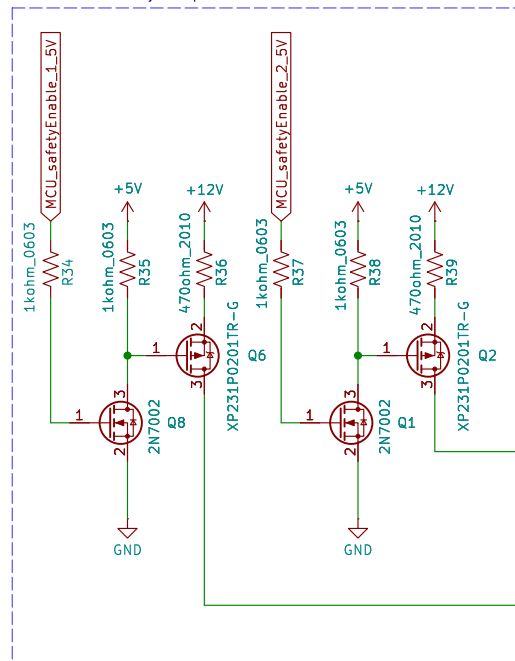
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Rev: 1

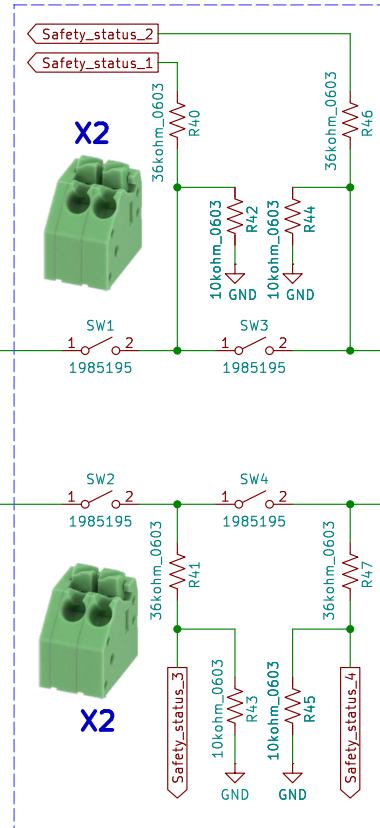
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Id: 5/9

two outputs from MCU
to enable safety loop



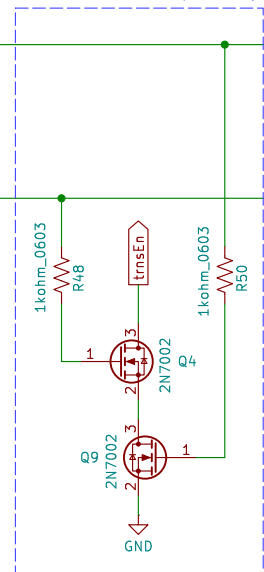
safety contacts for sensing if the lid is installed
feedback to MCU for each contact



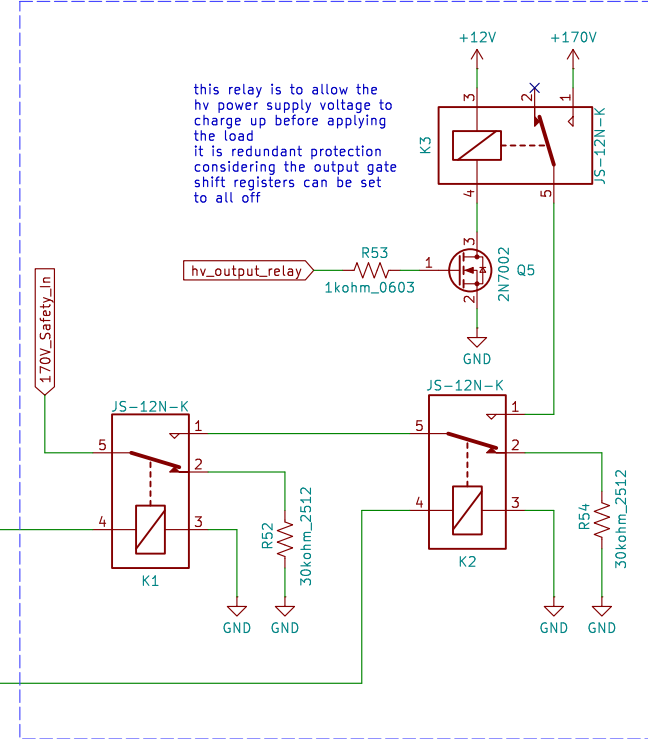
loop through
display PCB



enables HV PSU input relay



controls output of HV power supply



this relay is to allow the
hv power supply voltage to
charge up before applying
the load
it is redundant protection
considering the output gate
shift registers can be set
to all off

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Sheet: /Safety Circuit/

File: Safety Circuit.kicad_sch

Title: Safety Circuit – High Voltage PSU Control

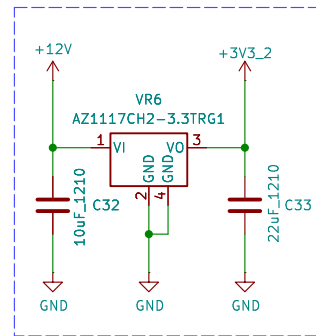
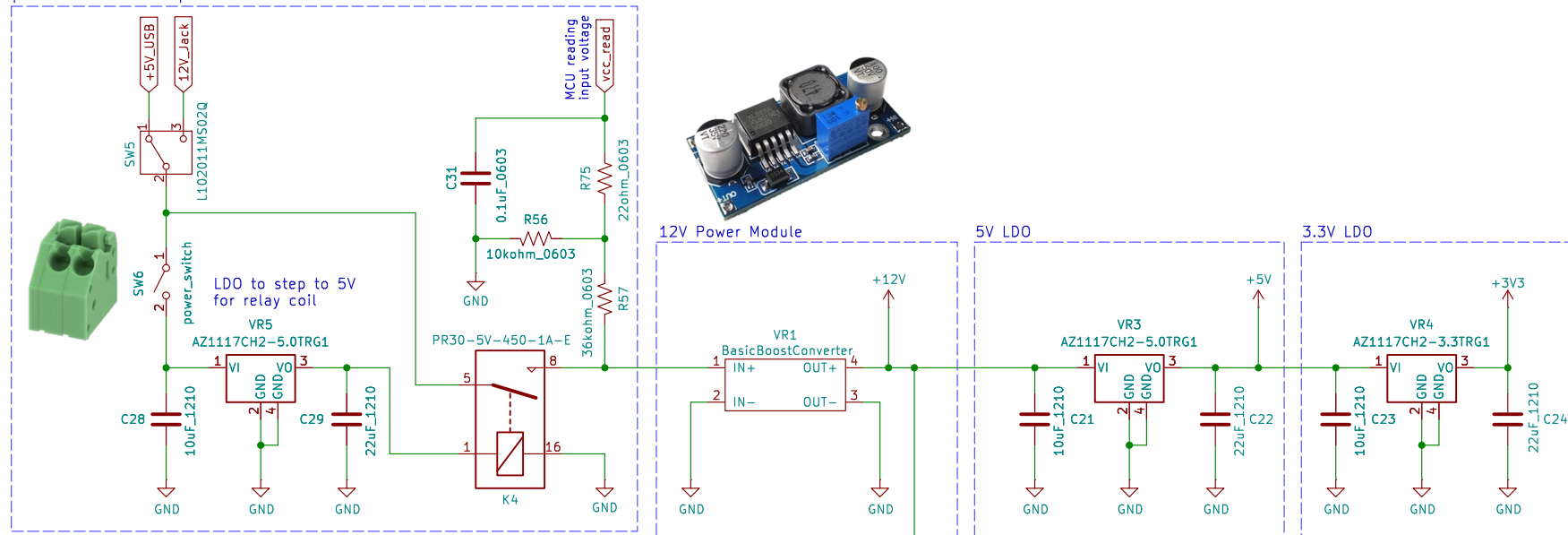
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Date: 2025-12-01

Rev: 1

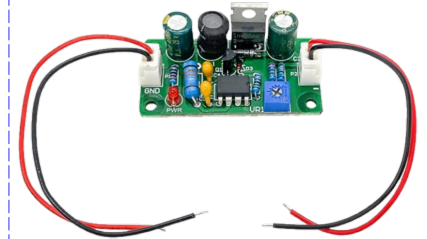
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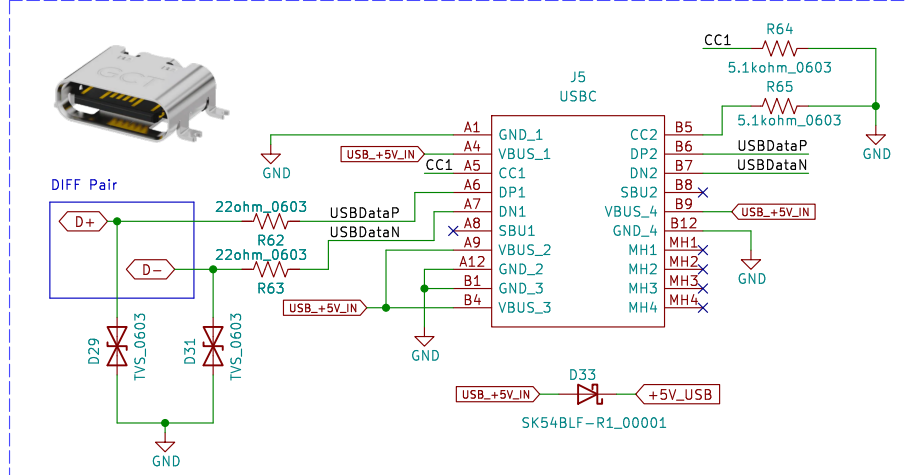
The schematic diagram illustrates the HV Power Supply circuit, divided into two main functional blocks:

- Controlled by safety circuit:** This section includes the input power stage. It features a +5V and +12V input. The +12V line passes through a 15kohm_0603 resistor (R58) and a 1kohm_0603 resistor (R59) before reaching the base of a PNP transistor (Q10, XP231P0201TR-G). The emitter of Q10 is connected to ground. The collector of Q10 is connected to the base of a MOSFET (K5, PR30-12V-450-1A-E). The MOSFET's source is connected to ground, and its drain is connected to the positive output of the HV_Power Supply (VR2).
- high voltage power supply:** This section contains the high-voltage generation and regulation stage. It starts with a transformer (T1, 860010578017) connected to the MOSFET's drain. The secondary winding of the transformer is connected to a series combination of two capacitors (C25 and C26) and a diode (D1, 1N4007) to ground. The output of this stage is connected to the positive input (IN+) of the HV_Power Supply (VR2). The HV_Power Supply (VR2) is a precision voltage source that regulates the output voltage. Its output (OUT+) is connected to the positive output of the HV_Power Supply. The output is also connected to a series combination of two capacitors (C30 and C31) and a diode (D2, 1N4007) to ground. The output voltage is monitored by a voltage divider (R73, R72, R55, R60, R74) connected to an HV_read input of the MCU. A 170V_Safety_In signal is also sent to the safety circuit.

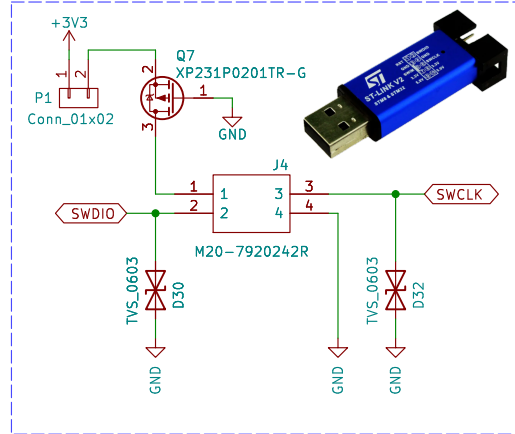


Rev: 1
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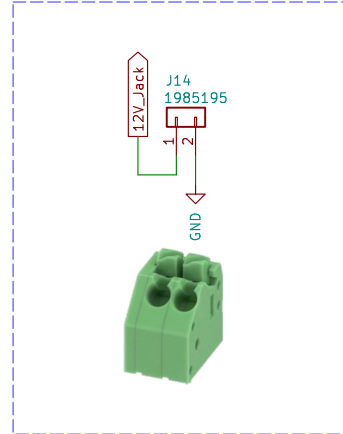
USB-C (2.0 Device Only)



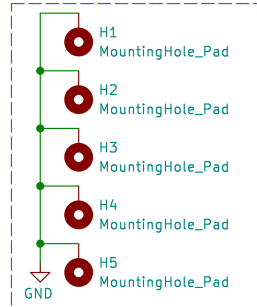
Programming Header



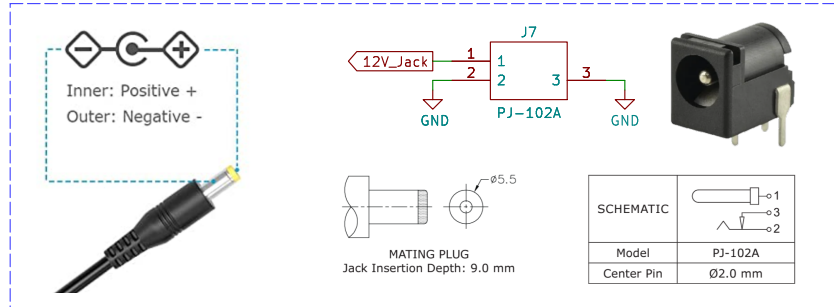
12VDC backup input



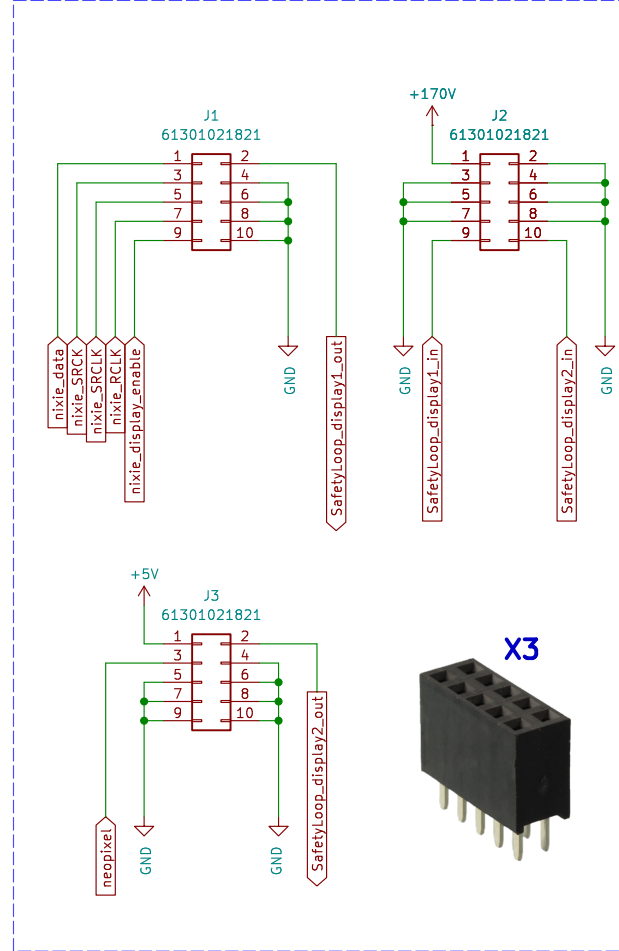
M3 mounting holes



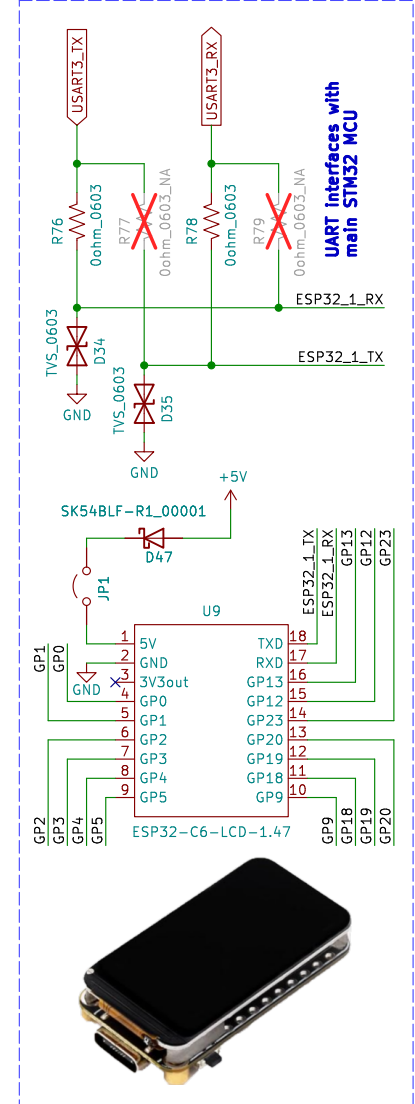
12VDC power input



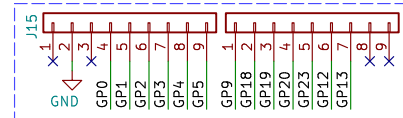
connections to driver board



ESP32-C6 1.47" 172x320 LCD



ESP32 Breakout Connections



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Sheet: /Connectors/

File: Connectors.kicad_sch

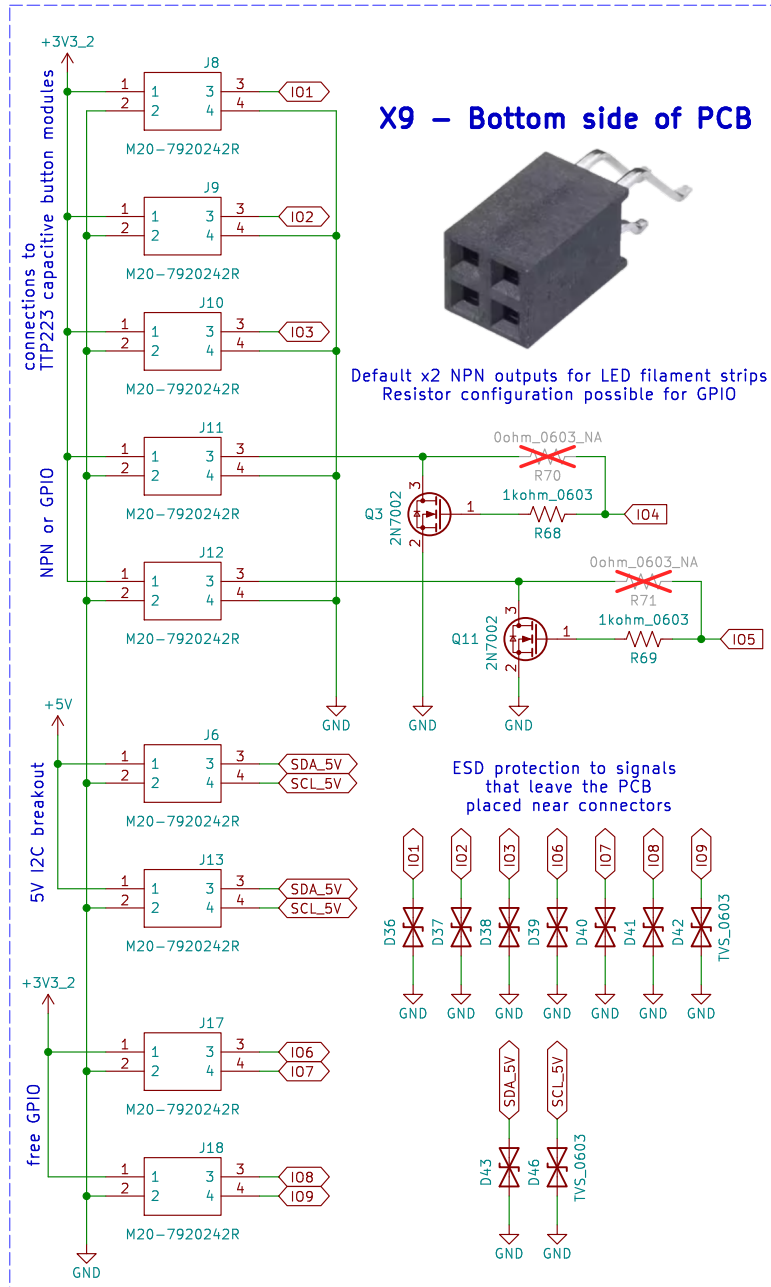
Title: Connectors

Size: A4 Date: 2025-12-01

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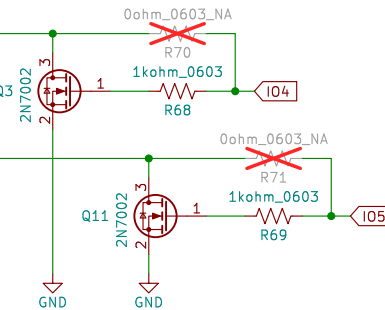
Rev: 1

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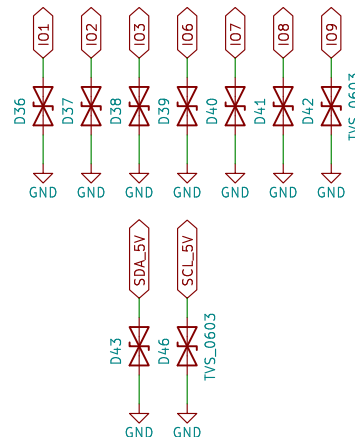


X9 – Bottom side of PCB

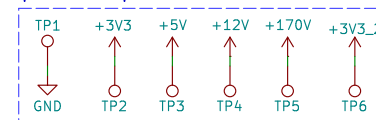
Default x2 NPN outputs for LED filament strips
Resistor configuration possible for GPIO



ESD protection to signals
that leave the PCB
placed near connectors



power test points



Sheet: /Connectors 2/
File: Connectors 2.kicad_sch

KiCad E.D.A. 9.0.6

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